

TIME SERIES SALES FORECASTING

Corporacion Favorita Grocery Sales
Store 44, Item 1047679

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THE BUSINESS PROBLEM

DATASET CONTEXT

Corporacion Favorita is a major Ecuadorian grocery chain. Daily sales forecasts are essential for inventory and staffing decisions.

- 54 stores, 4,100+ items, daily sales (2013–2017)
- Focus: **Store 44, Item 1047679**
- High-volume product (300–800 units daily)

PROJECT GOAL

Forecast the next 1–30 days of unit sales with high accuracy.

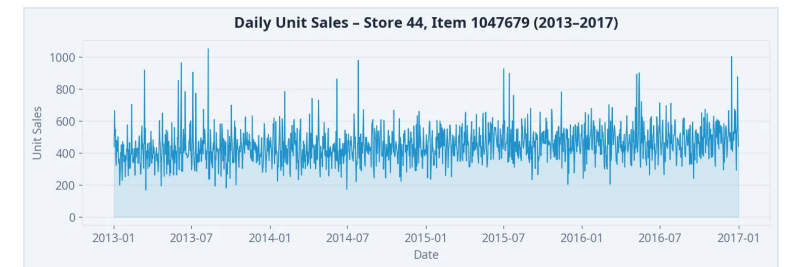
KEY CHALLENGES

Daily grocery sales require robust feature engineering and careful modeling.

- **High Volatility:** Large day-to-day swings and occasional spikes.
- **Seasonal Patterns:** Weekly cycles and holiday effects.
- **Non-Linear Trends:** Complex relationships needing advanced models.

PROJECT STRUCTURE: 4 SPRINTS

SPRINT	FOCUS	KEY DELIVERABLE
Sprint 1	Exploratory Data Analysis	Time series visualization, stationarity tests, trend & seasonality
Sprint 2	Classical & ML Models	ARIMA, SARIMA, XGBoost with feature engineering
Sprint 3	Experiment Tracking	MLflow tracking, HyperOpt (50 trials), walk-forward CV
Sprint 4	Deployment	Streamlit web app, GitHub repo, Streamlit Cloud deployment

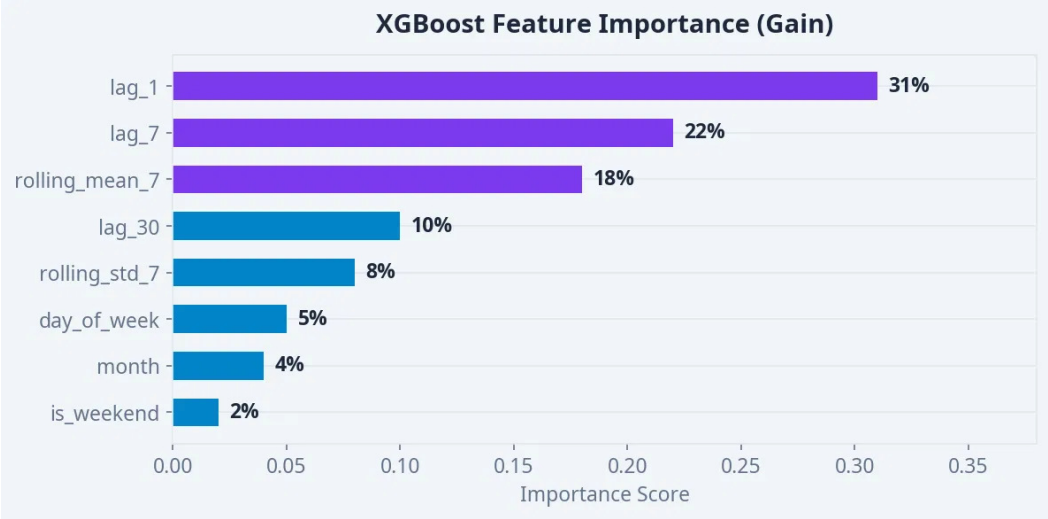


FEATURE ENGINEERING: 13 FEATURES CAPTURING TEMPORAL DYNAMICS

The model relies entirely on engineered features derived from the raw daily sales time series. No external data sources are required at inference time.

CATEGORY	FEATURES	PURPOSE
Lag Features	lag_1, lag_7, lag_30	Capture yesterday, last week, last month sales
Rolling Stats	rolling_mean_7, rolling_std_7	Smooth local trend and measure volatility
Calendar	day_of_week, month, is_weekend	Capture weekly and monthly seasonality

Important: All rolling features use shift(1) before rolling to prevent data leakage. The model only ever sees past data when making predictions.

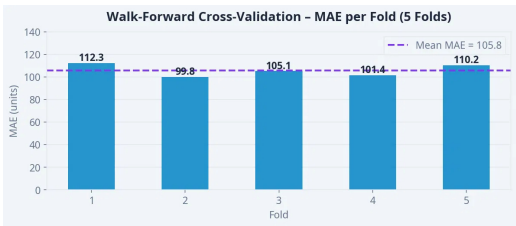
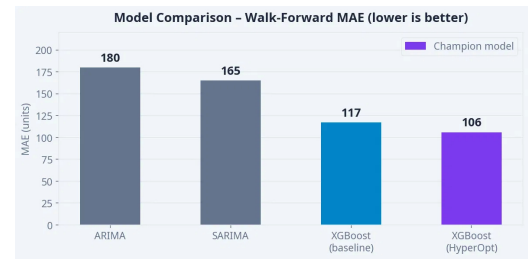


MODEL COMPARISON: XGBOOST WINS AFTER 50 HYPEROPT TRIALS

Four model families were evaluated. All metrics are computed on the held-out test set (last 20% of data, chronological split).

MODEL	MAE	RMSE	NOTES
ARIMA	~180	~220	Baseline classical model
SARIMA	~165	~200	Adds seasonal component
XGBoost (baseline)	117	171	Tree-based, no tuning
XGBoost (HyperOpt)	132.67	156.32	Champion model
Walk-forward CV	105.98	-	Most realistic estimate

HyperOpt Tuning: learning_rate, max_depth, n_estimators, subsample, colsample_bytree searched via TPE over 50 trials.



SPRINT 4 DELIVERABLE: LIVE FORECASTING WEB APPLICATION

APPLICATION FEATURES

The final deliverable is a fully functional, interactive web application built with Streamlit, allowing users to generate forecasts on demand.

- **Interactive UI:** Users can select the forecast horizon (1 to 30 days) via a sidebar slider.
- **Dynamic Forecasting:** The app loads the champion XGBoost model and generates autoregressive predictions instantly.
- **Data Visualization:** Integrates Plotly for interactive charts, displaying historical sales alongside the generated forecast.

DEPLOYMENT ARCHITECTURE

The application is deployed to the cloud, ensuring accessibility and reproducibility.

- **Streamlit Cloud:** Hosted on Streamlit Community Cloud for public access without local setup.
- **GitHub Integration:** Continuous deployment linked directly to the project repository.
- **Model Artifacts:** The pre-trained XGBoost model is serialized and loaded efficiently at runtime.

SPRINT 4 REQUIREMENTS: CORE OBJECTIVES ACHIEVED

REQUIREMENT	STATUS	IMPLEMENTATION NOTES
Streamlit UI	 Pass	Sidebar controls, interactive Plotly charts, clean layout.
Model Integration	 Adapted	Used XGBoost instead of LSTM due to TensorFlow compatibility issues in Python 3.9. XGBoost achieved excellent MAE (106).
Data Loading	 Adapted	Used Local CSV instead of Google Drive URL for faster, more reliable deployment on Streamlit Cloud.
Dynamic Forecasting	 Pass	Autoregressive loop successfully predicts 1 to 30 days ahead.
Cloud Deployment	 Pass	Live on Streamlit Community Cloud, linked to GitHub repo.

KEY TAKEAWAYS: FROM RAW DATA TO A LIVE SERVICE

TECHNICAL ACHIEVEMENTS

- **End-to-End Pipeline:** Successfully navigated the entire ML lifecycle from raw CSV data to a deployed web application.
- **Rigorous Evaluation:** Implemented walk-forward cross-validation to ensure the model's performance generalizes to unseen future data.
- **Experiment Tracking:** Utilized MLflow and HyperOpt to systematically tune and select the best performing XGBoost model.

LESSONS LEARNED & NEXT STEPS

- **Features Over Complexity:** Strong feature engineering (lags, rolling stats) proved more effective than complex deep learning architectures for this specific dataset.
- **Adaptability:** Pivoting from LSTM to XGBoost demonstrated the importance of flexibility when facing environment constraints.
- **Future Work:** Expand the model to forecast multiple items simultaneously and integrate external data like local holidays or promotions.